

Cloud Computing (COM712)

FHEQ Level: 7

Assessment: 3

Assessment weighting: 50%

Submission format: Report in PDF format, with optional ZIP folder for supporting artefacts

Submission Method: Via Brightspace upload

Submission Deadline: Monday 1 June 2026 by 12.00
midday (UK time)

Cloud infrastructure, networking and containerisation application

Required

This assessment tests your ability to design, deploy, and evaluate a multi-component cloud application using containers and orchestration tools. You will demonstrate that you can:

- Design and deploy a containerised cloud application using Kubernetes, Azure Container Instances, or AWS Fargate
- Configure scalable networking (VPC/VNet, ingress, service mesh) with security and monitoring
- Apply troubleshooting and problem-solving to performance or resilience issues
- Critically evaluate industry trends such as multi-cloud, GitOps, sustainability, and cost optimisation
- Communicate technical and reflective insight clearly to both specialist and non-specialist audiences

What you need to do

Select **one** of the three following scenarios:

Option A: Cloud-Shop Lite

Deploy a two-tier containerised web application (front end + database) on Azure Container Instances (ACI) or AWS Fargate. Use Docker Compose locally, then push images to ACR/ECR. Create a public IP, DNS name, and configure auto-scaling through ACI scale rules or Fargate auto-scaling. Test with a simple load-test script and document cost against a virtual-machine baseline.

Option B: Green-Compute Mini

Build a single-container CO₂ calculator that reads a CSV, processes data, and stores results in Azure Table Storage or DynamoDB. Deploy via Docker on ACI/Fargate, schedule automatic shutdown with Azure Logic Apps or AWS EventBridge, and compare carbon and cost performance with an always-on VM.

Option C: Your own innovation

Propose a unique containerised solution. Provide a short 250-word proposal outlining the concept, problem to be solved, and expected benefits. Your design must include at least two Docker containers, basic networking (VNet/VPC + public IP), storage, and a container runtime (ACI, Fargate, or local Docker Swarm). Kubernetes is optional.

Deliverables

You must submit a technical report in PDF format of around 1,500 words (±10%).

Optional items may also be included in a single ZIP folder:

- A demonstration video of two to four minutes showing your deployment, scaling, logs, or metrics
- A Git repository export or ZIP containing Dockerfiles, Kubernetes manifests, Helm charts, Terraform scripts, or similar artefacts
- A README file of up to 400 words describing quick-start steps, troubleshooting advice, and a cost and carbon estimate

If you choose to submit optional artefacts, name your folder **FamilyName_StudentID_A3.zip** and upload it with your report. If you submit only the report, it will be assessed on its own merits.

Report structure

Use the following headings in your report:

1. **Executive summary (≈ 100 words)**
Briefly summarise your design, key decisions, and outcomes.
2. **Scenario and requirements (≈ 150 words)**
Describe the business context, stakeholder goals, performance expectations, and constraints.
3. **Design justification (≈ 200 words)**
Compare ACI, Fargate, and VM options. Explain your chosen public-cloud environment and approach to auto-scaling or scheduled shutdown.
4. **Architecture diagram**
Provide a labelled diagram illustrating components such as Internet → Load balancer → Container service → Database → Storage. Include Azure or AWS service names.
5. **Build and deploy (≈ 300 words)**
Describe your process for containerisation, image management (Dockerfile → ACR/ECR), and deployment (ACI, Fargate, or Kubernetes). Discuss environment variables, secrets, and data-storage choices.
6. **Network, security, and scaling (≈ 250 words)**
Explain your network architecture and security measures (VNet/VPC, subnets, security groups). Discuss DNS configuration, scaling rules, and automated shutdown or load-balancing behaviour.
7. **Test, monitor, and fix (≈ 200 words)**
Summarise how you tested performance and resilience. Include metrics from monitoring tools such as CloudWatch or Azure Monitor, describe any faults or failures, and explain how you resolved them.
8. **Cost and carbon (≈ 150 words)**
Compare estimated seven-day running costs and carbon impact with a VM baseline. Reference the calculators or estimation tools used.
9. **Trends and next steps (≈ 150 words)**
Reflect on future developments in containerisation (for example serverless containers, green IT, or GitOps) and identify how your design could evolve.

Ethics reminder

You must not use any live personal data when completing this assessment. Use synthetic or publicly available data sets only to demonstrate testing, monitoring, or automation.

(Total – 100 marks)

Word count: Maximum 1,500 words (-/+10%) excluding reference list.

Learning outcomes assessed

- KLO 1: Demonstrate an in-depth knowledge of the essential concepts and principles underlying cloud computing
- KLO 2: Propose, develop, and evaluate a cloud-based solution for a real-world problem
- KLO 3: Demonstrate a critical awareness of current challenges and industry trends in cloud-based infrastructure
- SLO 1: Use relevant industry tools and services to perform configuration and deployment tasks
- SLO 2: Communicate technical work effectively to specialist and non-specialist audiences

Submission instructions

- Submit your report as a single PDF document via the Brightspace submission point for Assessment 3.
- Late submissions will be penalised according to policy.
- If you encounter any technical challenges, contact your tutor or learning support team well in advance of the deadline.

If you need any help accessing references, please contact knowledgebank@libf.ac.uk

KnowledgeBank, your e-library, is a collection of resources available to students and is accessible through <https://my.libf.ac.uk>

You should use this e-library to find additional books, journal articles and other information to support your studies. To begin your search:

- log in to <https://kb.libf.ac.uk> with your LIBF number and password;
- select 'Library Resources' from the link on the right of the home page.

Please see the next page for a table of suggested reputable sources that would be useful and is specific to this assessment/report task.

Marking Rubric for Cloud Computing (COM712) Assessment 3

Criteria	Fail (0-39%)	Marginal Fail (40-49%)	Pass (50-59%)	Merit (60-69%)	Distinction (70+)
Knowledge and understanding of cloud, container, and industry concepts (25%)	Major inaccuracies or omissions; no understanding of core container or orchestration concepts; no awareness of current industry trends.	Basic but incomplete knowledge; some confusion between container, network, or orchestration concepts; limited awareness of trends.	Sound understanding of containers, orchestration, and networking; adequate awareness of current developments.	Very good depth; clear links between theory and design decisions; accurate use of content from Weeks 6–8.	Outstanding depth and breadth; provides a nuanced critique of container technologies and industry direction; demonstrates original insight.
Critical thinking and problem-solving (design choices, trade-offs, troubleshooting) (25%)	Purely descriptive; lacks evaluation or reflection on technical decisions.	Limited critique; mainly lists features; minimal troubleshooting or reasoning.	Good evaluation of design choices (for example, Helm vs raw YAML, spot vs on-demand); some troubleshooting described.	Strong analysis of trade-offs such as cost, latency, or security; iterative fault-finding and reflection evident.	Sophisticated critique anticipating counter-arguments; uses metrics to defend redesigns; presents creative and well-reasoned solutions.
Application of evidence (deployment, monitoring, cost and sustainability metrics) (25%)	No evidence of a working configuration; screenshots or metrics missing; script non-functional; no cost estimate.	Weak or inconsistent evidence; partial deployment or incomplete cost analysis.	Adequate evidence showing all core services running; screenshots or metrics included; cost estimate provided.	Comprehensive and accurate evidence including performance and sustainability metrics; configuration reproducible with minor fixes.	Exceptional integration of theory and data; includes advanced metrics (for example, carbon impact, spot-price modelling, or SLA analysis); demonstrates high reproducibility.
Structure and communication (15%)	Disorganised or unclear; language and structure hinder understanding.	Structure or tone occasionally unclear; limited logical flow.	Well organised and readable; diagrams and figures understandable.	Very good flow and professional tone; visuals clear and well presented.	Exceptionally well-written, clear, engaging, and professionally presented; figures and visuals of excellent quality.
Referencing and academic conventions (10%)	No or poor referencing; serious accuracy issues; no reference to ethical practice.	Limited or inconsistent use of Harvard style; weak link to ethical or professional standards.	Mostly accurate Harvard referencing; at least ten credible sources; adequate mention of ethical considerations.	Accurate throughout; strong mix of academic and industry sources; clear ethical awareness.	Flawless Harvard referencing; integrates academic, technical, and professional sources; critical ethical reflection embedded.

Bonus uplift (within same class)

High-quality optional artefacts (video, Git repo, README) can raise mark **within** the awarded band but cannot jump bands (e.g., low merit → high merit).